

A Growth Obstruction to Free-Gibbs Barycenter Centering

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Source And Extracted Question

The source is Charles-Philippe Diez, *A sharp symmetrized free transport-entropy inequality for the semicircular law*, arXiv:2410.02715. In Remark 9 and the following conjecture, the paper asks whether, for every continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the one-dimensional free Gibbs growth condition

$$f(x) - 2 \log |x| \longrightarrow +\infty \quad (|x| \rightarrow \infty),$$

there exists $\lambda \in \mathbb{R}$ such that the free Gibbs measure associated with $f + \lambda \text{id}$ has barycenter zero.

Result

As stated with only the displayed logarithmic growth condition, the conjecture is false. The obstruction is simple: logarithmic confinement is stable under translations, but not under arbitrary linear tilts. A non-centered translate of a logarithmically confined potential gives an admissible potential for $\lambda = 0$, while every nonzero linear tilt destroys confinement in one direction.

Idea Of The Proof

Start with an even admissible potential V . By uniqueness, its free Gibbs measure is symmetric and has barycenter zero. Translate the potential: $f_a(x) = V(x - a)$. Translation invariance of the logarithmic energy makes the new equilibrium measure the translate of the old one, so its barycenter is a . If V grows only logarithmically, adding λx with $\lambda \neq 0$ makes the variational functional unbounded above by translating any fixed compactly supported trial measure far in the favorable direction. Thus $\lambda = 0$ gives the wrong barycenter and $\lambda \neq 0$ gives no free Gibbs measure.

Theorem 1 (Counterexample to the printed conjecture). *Let $c > 1$, $a \neq 0$, and*

$$V(x) = c \log(1 + x^2), \quad f_a(x) = V(x - a).$$

Then f_a is continuous and satisfies

$$f_a(x) - 2 \log |x| \rightarrow +\infty.$$

However, there is no $\lambda \in \mathbb{R}$ such that the free Gibbs measure associated with $f_a + \lambda x$ exists and has barycenter zero.

Proof. First,

$$f_a(x) - 2 \log |x| = c \log(1 + (x - a)^2) - 2 \log |x| = (2c - 2) \log |x| + o(\log |x|),$$

which tends to $+\infty$ because $c > 1$.

Let ν_V be the equilibrium measure for V , i.e. the unique maximizer of

$$\mu \mapsto \chi(\mu) - \int V \, d\mu.$$

Since V is even, uniqueness implies that ν_V is symmetric. Hence $\int x \, d\nu_V(x) = 0$. Let $\tau_a(x) = x + a$. The free entropy is invariant under translations, and

$$\int f_a \, d((\tau_a)_\# \mu) = \int V \, d\mu.$$

Therefore the unique maximizer for f_a is $\nu_{f_a} = (\tau_a)_\# \nu_V$. Its barycenter is a , so $\lambda = 0$ does not center it.

It remains to show that no nonzero λ is admissible. Let $\mu_0 = \nu_V$, and write $\mu_t = (\tau_t)_\# \mu_0$. The measure μ_0 is compactly supported under the standard one-dimensional logarithmic confinement hypothesis, and translation leaves χ invariant. For $\lambda > 0$,

$$\chi(\mu_t) - \int (f_a(x) + \lambda x) \, d\mu_t(x) = \chi(\mu_0) - \int V(y + t - a) \, d\mu_0(y) - \lambda \int (y + t) \, d\mu_0(y).$$

As $t \rightarrow -\infty$, the middle integral is $O(\log |t|)$, while $-\lambda t \rightarrow +\infty$. Thus the variational functional is unbounded above. Hence no maximizer, and therefore no free Gibbs measure, exists for $f_a + \lambda x$.

The case $\lambda < 0$ is identical, sending $t \rightarrow +\infty$. Consequently no real λ gives an existing centered free Gibbs measure. $\square$

Corrected Positive Statement

The obstruction above disappears if the potential is assumed to be superlinear in both directions. This is the natural hypothesis in the preceding argument of the source when linear tilts are introduced.

Proposition 1 (Superlinear repair). *Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and*

$$\frac{f(x)}{|x|} \longrightarrow +\infty \quad (|x| \rightarrow \infty),$$

and suppose that for every $\lambda \in \mathbb{R}$ the potential $f + \lambda \text{id}$ has a unique free Gibbs equilibrium measure ν_λ . Then there exists $\lambda_0 \in \mathbb{R}$ such that

$$\int x \, d\nu_{\lambda_0}(x) = 0.$$

Proof. Let

$$P(\lambda) = \eta(f + \lambda \text{id}) = \sup_{\mu} \left\{ \chi(\mu) - \int f \, d\mu - \lambda \int x \, d\mu(x) \right\}.$$

As a supremum of affine functions of λ , P is convex. The superlinear hypothesis implies coercivity: taking any compactly supported trial measure and translating it near a point where $-\lambda x - f(x)$ is large gives

$$P(\lambda) \geq C + \sup_x \{-\lambda x - f(x)\} - o(1),$$

and the right-hand side tends to $+\infty$ as $|\lambda| \rightarrow \infty$. Hence P attains a minimum at some λ_0 .

By the standard envelope theorem for a convex supremum with a unique maximizer (equivalently, Danskin's theorem),

$$P'(\lambda) = - \int x \, d\nu_\lambda(x)$$

at every λ . Since λ_0 minimizes P , $P'(\lambda_0) = 0$, which gives the desired barycenter identity. $\square$

Verification And Limitations

The counterexample uses only standard facts recalled in the source: existence and uniqueness of the one-dimensional equilibrium measure under the logarithmic growth hypothesis, uniqueness of the free Gibbs maximizer, and translation invariance of the logarithmic energy/free entropy.

The negative result is a literal reading of the printed conjecture under $f(x) - 2 \log |x| \rightarrow +\infty$. It does not refute the intended centering statement when f is assumed to grow faster than every linear function, or more simply when $f(x)/|x| \rightarrow +\infty$. The proposition records that corrected form.

Novelty Check

The run indexes were searched for arXiv:2410.02715 and for phrases involving free Gibbs measures, barycenter zero, linear tilts, free pressure, and Blaschke–Santalo. No existing run packet was found. Web searches on 2026-07-03 for exact phrases such as “free Gibbs measure barycenter zero $f + \lambda$ ” and “free Blaschke–Santalo barycenter zero lambda free Gibbs” found the source paper but no separate answer.

References

- C.-P. Diez, *A sharp symmetrized free transport-entropy inequality for the semicircular law*, arXiv:2410.02715.
- E. B. Saff and V. Totik, *Logarithmic Potentials with External Fields*, Springer, 1997. Used through the standard one-dimensional equilibrium existence and uniqueness facts cited in the source.
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